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Effect of an AI-based chatbot on students' learning performance in alternate reality game-based museum learning

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Abstract

Recently, alternate reality games (ARGs) have been applied in museum learning to encourage learners' active engagement through playful problem-solving activities. However, most learners have insufficient prior knowledge and metacognitive skills to complete the learning tasks in such games. To support learning with ARGs, there is a need to provide proper feedback so that the learners are capable of self-regulated learning and solving the problems encountered during the learning process. Considering the influences of individual differences, this research developed a chatbot as a learning partner in ARG-based learning to support students' learning with adaptive feedback in a museum. A quasi-experimental study was conducted to evaluate the effectiveness with and without the proposed approach. The results showed that the AI-based chatbot approach could significantly improve learners' metacognition awareness, emotional engagement and behavioural engagement. Besides, it is helpful to facilitate students' double-loop learning.

KEYWORDS

alternate reality games, chatbots, computer-mediated feedback, game-based learning, museum learning

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Practitioner notes

What is already known about this topic

- Alternate reality games (ARGs) enable students to interact with real-world contexts.
- The complexities of real contexts and ambiguities of the ARGs could frustrate students and demotivate their engagement in problem-solving activities.
- Conventional corrected feedback may fail to match individual students' requirements in solving encountered problems.

What this paper adds

- An AI-based chatbot was designed as a smart learning partner to support the students in solving encountered problems.
- The experimental results showed that the proposed approach improved students' metacognitive awareness and engagement.
- A drawing analysis and a behaviour sequential analysis were applied to further examine the students' perceptions and behavioural patterns.

Implications for practice and/or policy

- It is potentially worth integrating AI-based chatbots into game-based learning to promote students' metacognitive awareness and engagement.
- AI-based chatbots provide adaptive feedback to effectively facilitate students' self-regulated learning and double-loop learning.

INTRODUCTION

Museums provide a free-choice learning environment with fruitful interdisciplinary learning resources to support competency-based learning in authentic learning contexts (Valanne et al., 2017). Considering that museum learning is a dynamic process, and considering the results of the interactions between students and the museum context, students' active engagement is important to promote their learning in museums (Falk & Dierking, 2018). Recently, alternate reality games (ARG) have been widely applied to promote students' active engagement and motivation in museums (Jerrett et al., 2017; Meekaew & Ketpichainarong, 2021). ARG is defined as a new form of collaborative game-based learning which engages students in interaction with both virtual and real worlds with a series of problem-solving activities and mastery narratives (Whitton et al., 2014). However, researchers have indicated that ARGs benefit learning only when students can complete the complex challenges in the ARGs; otherwise, ARGs may negatively affect students' learning (Connolly et al., 2011). Previous research explained that many students have insufficient metacognitive abilities to apply proper learning strategies to solve encountered problems in open-ended museum contexts (De Beer & Bothma, 2016). As a result, students were demotivated by frustrating learning experiences (Whitton et al., 2014).

To promote students' learning with ARGs in museums, it is essential to provide proper learning support to improve their metacognitive abilities and help them solve problems encountered during their learning process. To this end, previous research reported that chatbots effectively supported diverse students' learning by providing adaptive feedback according to individual requirements, which is helpful for improving students' metacognitive skills in problem-solving activities (Chang et al., 2022). Besides, by providing a natural language-based interface, AI-based chatbots have great potential to facilitate students' social interactions and motivation, which promotes their active engagement in learning (Fryer

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et al., 2019; Tao et al., 2021). AI-based chatbots effectively support students in solving problems with immediate feedback and supplementary information, which can be seen as a personal tutor or assistant for students (Lo, 2023; Sok & Heng, 2023). AI-based chatbots effectively advance teaching and learning by providing adaptive learning support and assessment to reflect on individual students' learning progress and achieve a more interactive learning environment (Grassini, 2023). In this vein, this study applied a chatbot as a learning partner in an ARG to improve students' metacognitive abilities and engagement. The chatbot was expected to support students in solving problems encountered in the ARG. Furthermore, more attention was paid to the students' learning achievement, metacognitive awareness, cognitive engagement, emotional engagement and learning behaviour engagement. The research questions are as follows:

1. Do the students who learn with the AI-based chatbot (AIC-ARG) approach have better learning achievement than those who learn with the conventional ARG (C-ARG)?
2. Do the students who learn with the AIC-ARG have better metacognitive awareness than those who learn with the C-ARG?
3. What are the differences in the cognitive engagement of the students who learn with the AIC-ARG and those who learn with the C-ARG?
4. Do the students who learn with the AIC-ARG have better emotional engagement than those who learn with the C-ARG?
5. What are the differences in the behavioural engagement of students who learn with the AIC-ARG and those who learn with the C-ARG?

LITERATURE REVIEW

Alternate reality games (ARGs)

ARGs engage students in interacting with both virtual and real worlds by using existing digital technologies and problem-solving activities (Elsom et al., 2021). With their collaborative problem-solving activities, many studies have found that ARGs are helpful in enhancing students' knowledge interpretations and their high-order cognitive abilities such as critical thinking, problem-solving and collaboration skills (Gilliam et al., 2017; Tulloch et al., 2021). ARGs create an immersive museum interactive experience, encouraging students' active engagement with museum exhibitions (Berge & Muilenburg, 2013). Previous research found that ARGs effectively motivate students to interpret the related knowledge of exhibits and keenly observe the exhibit details (Hsu & Liang, 2022). Besides, ARGs facilitate meaningful social interactions among students to solve problems (Koepfler et al., 2017). However, the ambiguity of ARGs and the complexity of real-world contexts increase the challenges for students to solve the encountered problems (De Beer & Bothma, 2016). Previous research found that many students hardly finished the ARG learning tasks, and in turn, the frustrated students became disengaged and had difficulty achieving the expected learning outcomes (Whitton et al., 2014).

AI-based chatbots

AI-based chatbots have been widely applied in education (Kohnke, 2022; Pérez et al., 2020). AI-based chatbots can be smart learning partners or assistants which can effectively support students' learning (Fryer et al., 2019; Lin & Mubarok, 2021). According to the zone

of proximal development theory, students can learn from social interactions with chatbots in such authentic conversation contexts (Vygotsky, 1978). Chatbots are interactive tools that provide supplementary information and immediate feedback to students to help them solve encountered problems (Huang et al., 2021; Yin et al., 2021). Besides, AI-based chatbots provide corrective feedback for learning evaluation, which is helpful for students to modify their learning actions and promote self-regulated learning (Bibauw et al., 2019), as well as improving their learning performance (Smutny & Schreiberova, 2020), learning engagement and motivation (Hew et al., 2023). Researchers have also found that chatbots facilitated social connection within collaborative learning groups, which benefited conducting meaningful social interactions and improved collaborative learning (Yilmaz & Yilmaz, 2019).

Despite the fruitful literature demonstrating the positive effect of AI-based chatbots on learning, there is a lack of evidence to identify the effects on museum learning. Some museums might provide chatbots for guided tour information and visiting path planning; however, few studies have employed chatbots for educational purposes in museum contexts (Sperlí, 2021). Therefore, this study applied an AI-based chatbot in an ARG and investigated the effects on students' learning achievements, metacognitive awareness and engagement in museum contexts.

AI-BASED CHATBOT IN ALTERNATE REALITY GAME FOR MUSEUM LEARNING

System framework and interface

As shown in Figure 1, a smart learning partner named Bubi was implemented in an ARG system to support students' active learning in a museum. The ARG system consists of a learning content management sub-system (LCMS), a chatbot management sub-system and a learner management sub-system (LMS). The LCMS manages the learning content library, which includes exploring maps, game quests, learning summaries and rewards for each ARG quest. Besides, it manages the conventional ARG's feedback including no-response feedback and corrective feedback with different levels of detail according to the wrong answer times. The chatbot management sub-system managed the AI-based chatbot content to support individual students' self-regulated learning in the museum, which includes the questioning prompt list, learning diagnosis and supportive feedback. The LMS manages the game process and learning portfolios. The game process records the students' game progress status and the results of each quest. The learning portfolio records students' learning behaviours, including their gaming logs, time and answer results of each quest (as correct or not). The LMS is also linked to the museum membership management system so that the students' learning portfolios can be updated to promote return visits.

The students are asked to learn with Bubi to complete all the ARG quests. Figure 2 displays the ARG system flow. In the beginning, the students view the quest locations and plan their learning path in the exploring area. A game story is provided to explain the quest goal before each quest starts. After they select a quest, the students collect related information from the museum exhibitions and make annotations to solve the quests. Once they submit the correct answer to solve the quest, a learning summary is provided to summarize the learning concepts of the quest. On the other hand, Bubi gives the students corrective feedback when they give wrong answers.

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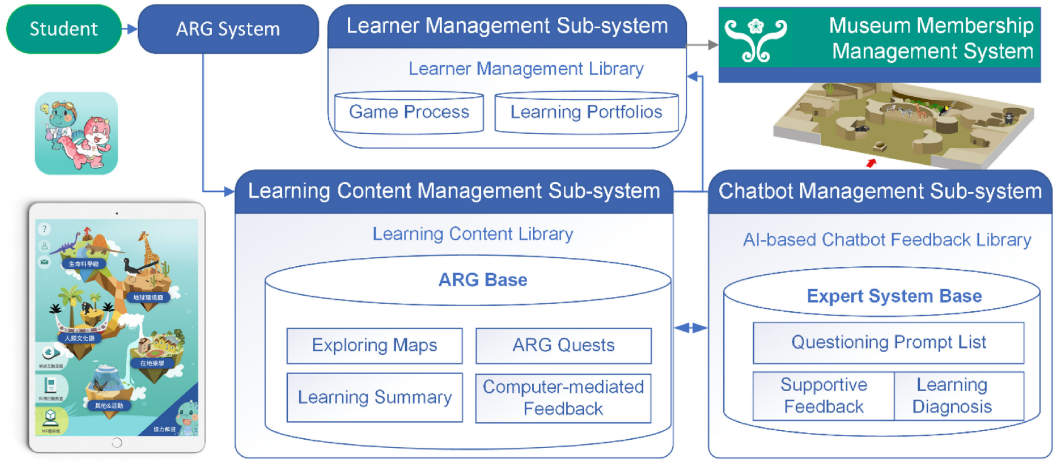


FIGURE 1 The system framework of the AI-based chatbot approach in the ARG system.

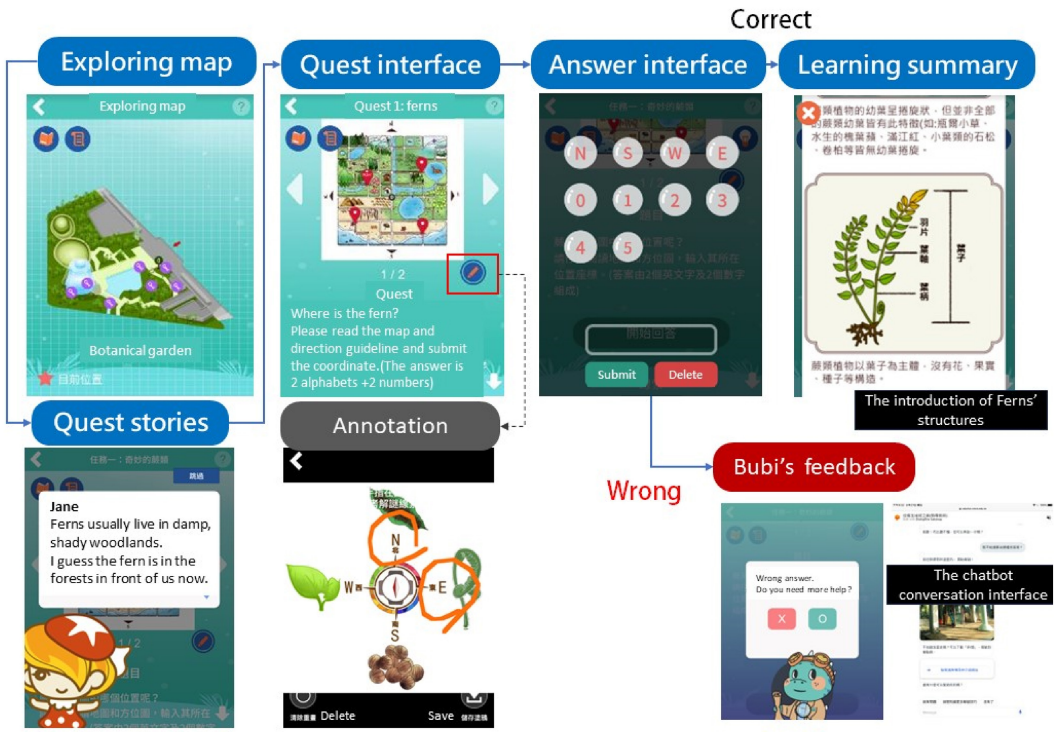


FIGURE 2 The system flow of the ARG system.

The conventional ARG's feedback

Considering that the early iteration of elaborated feedback may negatively affect students' reflection on learning to develop problem-solving abilities, Bubi gives the students feedback with different levels of detail according to the number of wrong answers they give (Deeva

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et al., 2021). The first time the learner gives a wrong answer, Bubi gives simple feedback with answer results (right or wrong). For more wrong answers, Bubi gives elaborative feedback to guide students in solving the encountered problems. Considering that most students have limited patience to read text, Bubi asks the students to revise the information carefully after they submit their second wrong answer. The feedback of the first and second wrong answers is shown in Figure 3.

For the third wrong answer, Bubi gives the students a step-by-step guide so that they can refer to the steps to solve the problem (Figure 4).

For the fourth or more wrong answer, Bubi gives the students metacognitive feedback which highlights the common mistakes to help them revise and correct their mistakes to solve the problem. Regarding no-response feedback, Bubi actively shows the quest's clues if the students do not have any action for over 120 seconds. Besides, the students can review the step-by-step guide by clicking the question-mark button after the third wrong answer.

AI-based chatbot approach design

At the first wrong answer, Bubi gives simple feedback to avoid the negative effect of early iterations of elaborated feedback (the C-ARG's feedback). After the second wrong answer, Bubi asks the students if they need help, and a chatbot is used to provide the students with adaptive feedback based on their specific requirements (Figure 5).

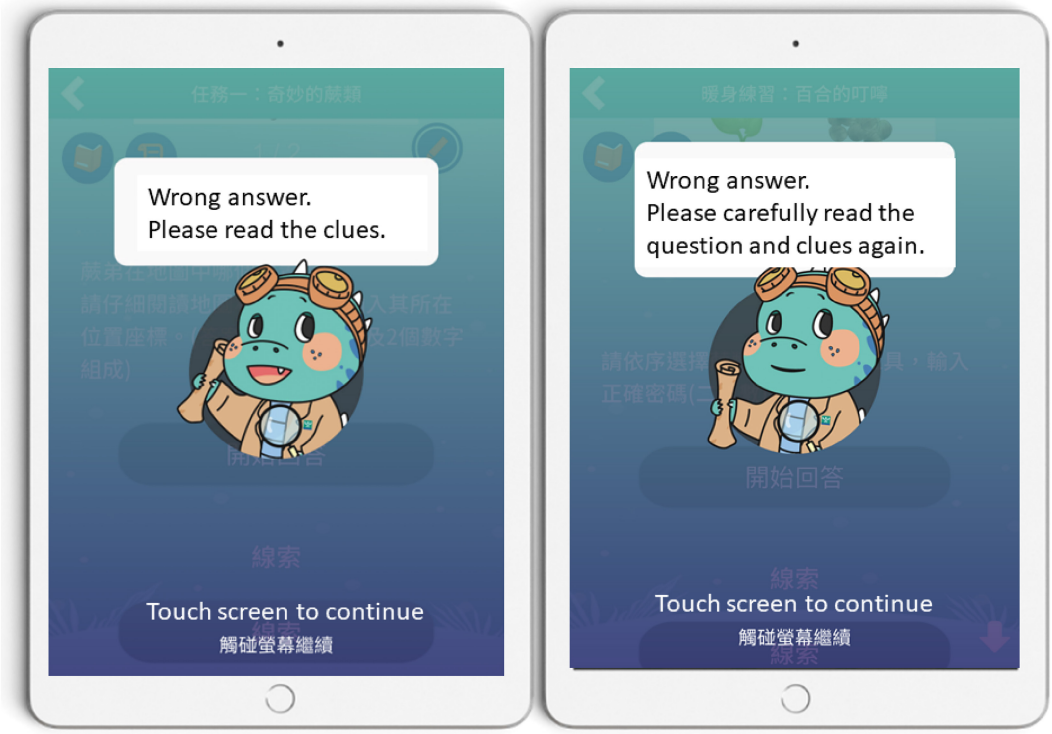


FIGURE 3 The C-ARG's feedback on the first (left) and second (right) wrong answer.

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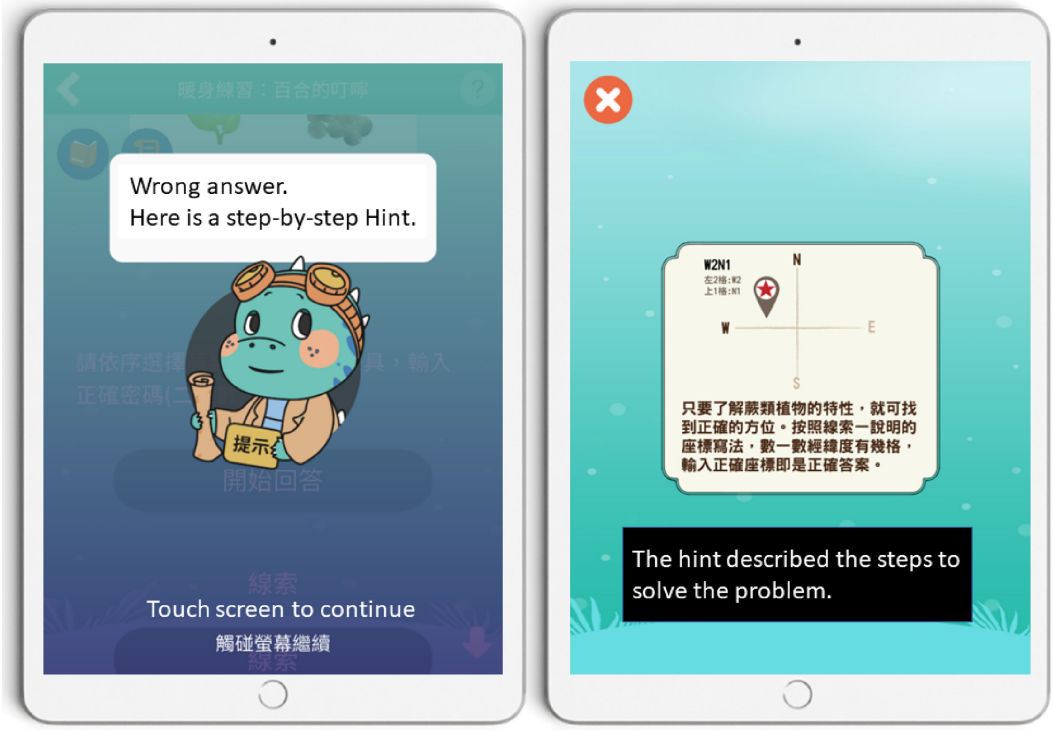


FIGURE 4 The C-ARG's feedback on the third wrong answer.

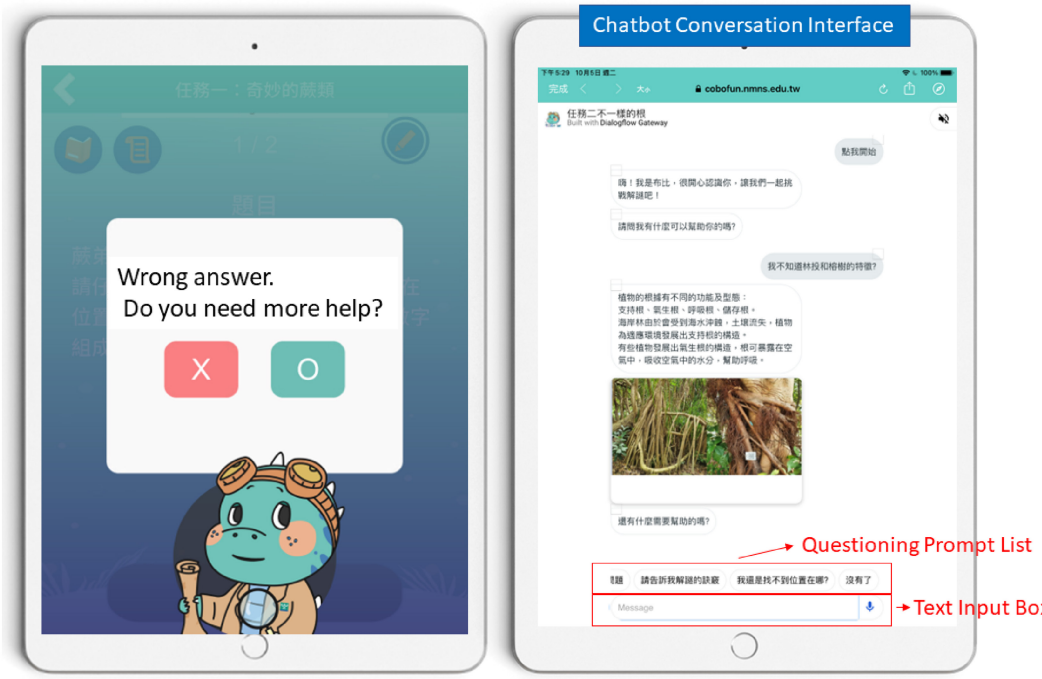


FIGURE 5 The AI-based chatbot's feedback on the second and more wrong answers.

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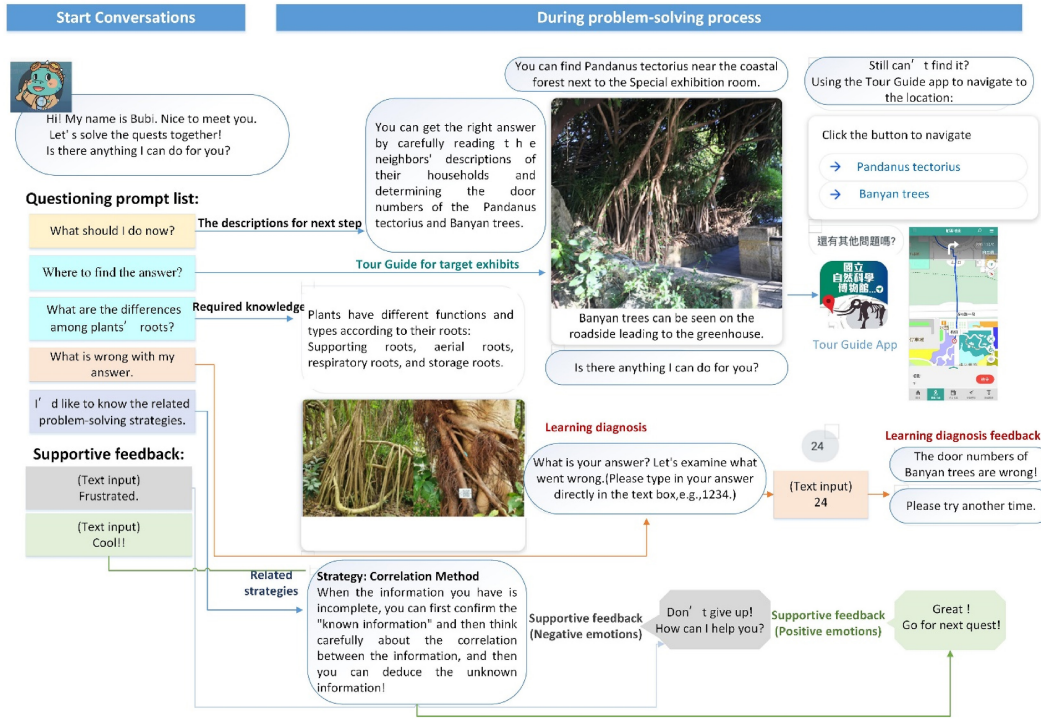


FIGURE 6 An example of dialogue scenarios with the AI-based chatbot.

Considering the negative effect of insufficient prior knowledge, experiences and meta-cognitive abilities on learning, adaptive feedback applies a meta-design approach to provide the students with proper support for dealing with the possible difficulties during their gaming process (Fischer & Giaccardi, 2006). The students can use a questioning prompt list to find the supportive information they need, or they can directly talk to Bubi with text messages (Figure 6). There are three types of feedback:

1. Questioning prompt list:

The questioning prompt list provides a list of questions to help the students satisfy their individual information needs. For insufficient prior knowledge and visiting experience, Bubi gives related information and a navigation map to help students build the required knowledge and navigate to the target exhibit. For insufficient metacognitive abilities, Bubi gives the students supportive information to help them identify the problem and plan their actions to solve the problem, such as tips for what to do next and the recommended problem-solving strategies for solving specific types of problems.

2. Learning diagnosis:

The learning diagnosis feedback is provided to support students in reflecting on their learning results. After the students submit their answers, Bubi gives the students feedback to identify the possible mistakes and promote their self-regulated learning to solve the problem.

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3. Supportive feedback:

To maintain students' learning motivation and to improve their affective experiences, Bubi gives supportive feedback according to their affective responses. For positive affective responses, Bubi gives praise and encouragement to encourage the students; for negative affective responses, Bubi encourages the students to keep trying and also gives the students a questioning prompt list to guide them to continue solving problems.

On the third wrong answer, the C-ARG's feedback is also given to explain the sequences to solve the problem. Besides, Bubi actively talks to students if they do not have any action for over 120 seconds, and the students can ask Bubi for adaptive feedback at any time by clicking the question-mark button after the third wrong answer.

EXPERIMENTAL DESIGN

Learning scenarios

A learning activity was designed to support students' learning in the museum. An ARG named ‘Lulu's Birthday Party’ was used to guide students in learning knowledge about the diversity of plants in the National Museum of Natural Sciences in Taiwan. The game included a warm-up quest for gameplay instructions and six quests to guide the students to explore the museum exhibitions to learn the characteristics of the plants in different environments, such as the waterfront or tropical rainforest plants. Each quest presents a learning concept with different exhibits to learn about the diverse plants. The students are asked to collaborate with their teammates and solve all the quests together.

Participants

A learning activity was conducted at the National Museum of Natural Science in Taichung, Taiwan, to examine how AI-based chatbots affected the students' learning in the museum. Two sixth-grade classes from different elementary schools participated in this activity. A total of 17 students in one class were assigned to the experimental group and were asked to learn with the AIC-ARG, while 22 students in another class were assigned to the control group and were asked to learn with the C-ARG. The teachers had received consent from students and their guardians before registering for the activity. Besides, they confirmed that all students had basic abilities to manipulate tablets independently, and had learned the related knowledge in school; however, they did not have ARG gaming experience, nor any experience of visiting the museum.

Measurements

Learning achievement

A pretest and posttest were used to examine students' learning achievement. The pretest included five multiple-choice questions (50%) and five true-or-false questions (50%) to evaluate students' prior knowledge regarding the diversity of plants, that is, the characteristics of plants in tropical rainforests, coastal forests and river shoals. For example, what is the root structure of a banyan tree that helps it absorb moisture from the air and breathe? The

posttest included six multiple-choice questions (60%) and nine matching questions (40%) to evaluate students' learned knowledge regarding the diversity of plants after the learning activity. Both tests were designed by experienced teachers from the compulsory education advisory group of Taichung City. The KR20 values of the pretest and posttest were 0.51 and 0.82, respectively, which represented acceptable reliability of the tests (Saidi & Siew, 2019).

Metacognitive awareness

Metacognitive awareness is defined as the key metacognitive ability to use reflective thinking to be aware of one's own status during the learning process in a given context (Ridley et al., 1992). The metacognitive awareness pre- and post-questionnaires (Cronbach $\alpha = 0.94$) were modified from the Metacognitive Awareness Inventory (Schraw & Dennison, 1994), which consists of nine questions with Likert-type scales ranging from 1 (*strongly disagree*) to 5 (*strongly agree*) for measuring the changes in metacognitive abilities after the learning activity, for example, I use different strategies depending on the situation.

Cognitive engagement

Cognitive engagement is defined as one's psychological investment in learning (Huang et al., 2018). The drawing analysis technique was used to examine the students' cognitive focus during the learning activities to identify what they focused their efforts on. The students were asked to draw their learning impressions of this activity and to describe what they drew with three open-ended questions, that is, 'What did I do today?', 'How did I feel about the activity?' and 'What did I learn today?'. The drawing analysis coding scheme was modified from the coding scheme of Lai (2021), which examined students' perceptions according to five aspects: participants, learning environment, learning activities, objects and emotions. Table 1 displays the drawing coding scheme.

Before coding, the researcher explained the coding method and coding scheme and randomly selected one drawing to demonstrate the coding method. The two coders discussed reaching agreement on each code before they coded each of the 39 drawings, and they coded all the drawings from the drawing content and descriptions (Kappa = 0.88). An example of a coded drawing is shown in Figure 7:

Emotional engagement

Emotional engagement is defined as one's subjective perceptions, attitude, value and interest in learning (Lee et al., 2022). The emotional engagement questionnaire (Cronbach

TABLE 1 Drawing coding scheme of cognitive engagement.

Distributions	Items
Participants	Teacher, Student, Bubi, Others and No participant
Environment	Greenhouse, Outdoor, Waterfront and Others
Activity	Observation, Collaboration, Problem solving, Manipulating Tablets, Exploration and Others
Objects	Plants, Plant details, Creatures, Tablets, Landscape and Others
Emotions	Joy, Sadness, Confusion, Others and No expression

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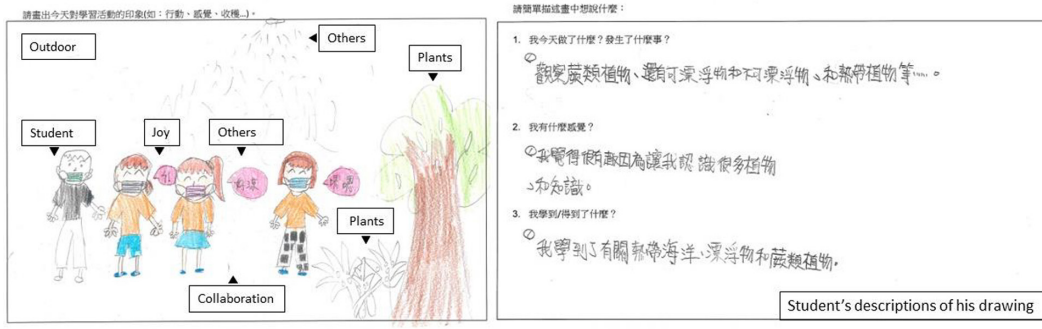


FIGURE 7 An example of a coded drawing (A19).

TABLE 2 Game behaviour coding scheme.

Code	Behaviour	Descriptions
K	Skip the story	Click the skip button to skip the quest story
Q	Start a quest	Click the explore map quest button to start a game quest
F	Call for Bubi	Click the question-mark button or have no actions for over 120 seconds, so Bubi actively gives the student feedback
H	Step-by-step guide	Seek help
A	Chat with Bubi	Open the conversation window to chat with the AI-based chatbot
P	Draw a note	Use the drawing function to take notes
S	Submit an answer	Click the submit button to start answering a quest
W	Wrong answer	Submit a wrong answer
C	Correct answer	Submit a correct answer
L	Read quest clues	Click the clue button to read the supplementary information related to the quest
N	Read learning summaries	Read the learning summaries after completing a quest
M	Explore map	Stay on the exploring map interface

$\alpha = 0.79$) was modified from the game engagement questionnaire (Kim et al., 2017), which includes four questions with Likert-type scales ranging from 1 (*strongly disagree*) to 5 (*strongly agree*) for measuring their subjective perceptions in learning, for example, I enjoyed the learning activity a lot.

Behavioural engagement

Behavioural engagement is defined as students' involvement in learning activities (Huang et al., 2018). Each student's learning behaviours were recorded in system logs to evaluate their behavioural engagement to identify their engaged time and problem-solving strategies. Their behavioural engagement was evaluated from their stay time for each exhibit, learning behaviours and interactions with AI-based chatbots. The stay time for each exhibit refers to their spending time on each quest. Considering the negative effect of unequal participation in collaborative learning, spend time of less than 60 seconds was defined as free-riding behaviour to identify individual participation in collaborative learning.

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The students' learning behaviours of ARGs were coded and analysed with a lag sequential analysis to compare the differences between the problem-solving patterns of the experimental and control groups. As shown in Table 2, the game behaviour coding scheme was modified from Yang's (2017) game behavioural coding scheme.

Procedure

In the beginning, a 10-minute introduction was given to explain the procedure of the learning activity. After the students completed the pretest and metacognitive awareness pre-questionnaire (20 minutes), they were asked to collaboratively learn in groups of three or four to solve the ARG, 'Lulu's Birthday Party', together within 60 minutes. Each student used an iPad mini to play the game. Regarding the experimental group, the students learned with the AIC-ARG and communicated with Bubi and their teammates to solve the quests together. Conversely, the students in the control groups had fewer bidirectional interactions with Bubi. The students learned with the C-ARG and mainly collaborated with their teammates to solve problems; Bubi gave them feedback when they gave wrong answers.

All students' learning behaviours regarding the system manipulation and stay time during the gaming process were recorded for behavioural engagement analysis. After they completed the ARG, they were required to fill out the posttest and questionnaires to evaluate their learning achievement, metacognitive awareness and emotional engagement (20 minutes). Besides, they were asked to draw their learning impressions and answer the open-ended questions to examine their cognitive engagement and perceptions (20 minutes). The total activity duration was about 130 minutes.

RESULTS

Learning achievements

A one-way ANCOVA was performed after confirming the assumption of homogeneity of regression ($F_{(2)} = 0.38$, $p > 0.05$) to compare the learning achievements of the experimental group and the control group. As shown in Table 3, the students who learned with the AIC-ARG had similar performances to those who learned with the C-ARG ($F = 0.11$, $p > 0.05$).

TABLE 3 The ANCOVA results of learning achievements.

Group	*N*	Mean	*SD*	Adjusted mean	*SE*	*F*	η^2
Experimental	17	67.82	19.06	68.10	5.32	0.11	0.00
Control	22	66.00	23.37	65.79	4.67		

TABLE 4 The Mann–Whitney *U* test results of metacognitive awareness.

Variables	Group	*N*	Mean	J-T	*U*	*z*	p^2	*r*
Pre-questionnaire	Experimental	17	4.24	0.52	157.50	−0.84	0.40	0.13
	Control	22	3.94	0.88				
Post-questionnaire	Experimental	17	4.40	0.57	92.00	−2.97**	0.01	0.48
	Control	22	3.66	0.90				

**$p < 0.01$.

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Metacognitive awareness

Levene's test of determining homogeneity of variance was violated ($F = 22.02$, $p > 0.05$), indicating that the null hypothesis was rejected. Therefore, a Mann–Whitney U test was conducted. Table 4 displays the results of students' metacognitive awareness before and after the learning activity. The results showed that there were no significant differences between the two groups' metacognitive awareness before the learning activity, while the students who learned with the AIC-ARG significantly outperformed those who learned with the C-ARG ($U = 92.00$, $p < 0.01$, $r = 0.48$).

Cognitive engagement

To examine the students' cognitive engagement in the two approaches, Chi-square tests were used to compare the students' learning impressions in different aspects. The results showed that there were no significant differences in participants, environment or activity ($p > 0.05$), while the students who learned with different approaches had a significantly different focus on objects and emotions ($p < 0.05$). Table 5 shows the students' attention to objects. Students who learned with the AIC-ARG paid more attention to drawing the scenes of learning activities, such as sun, clouds and buildings. Besides, some students used four-panel manga to describe their learning process (Figure 8).

Students who learned with the C-ARG described more plant details in their paintings compared to those in the experimental group (Figure 9). In general, most students in the experimental group focused on describing what they did in the learning activities, while those in the control group focused more on what they saw in the learning activities.

As Table 6 shows, there is a significant difference in students' main emotions in the learning activities. The students in the control group mainly showed positive emotions, while those in the experimental group showed more complex feelings about the learning activities, such as being confused or surprised.

Emotional engagement

A Mann–Whitney U test was conducted to examine the students' emotional engagement of the two approaches (Table 7). The results show that the students who learned with the AIC-ARG had significantly higher emotional engagement than those who learned with the C-ARG.

TABLE 5 The students' attention to objects in the learning activities.

Objects categories	Experimental group (intergroup ratio)	Control group (intergroup ratio)	Total (total ratio)
Plants	20 (29.4%)	20 (23.0%)	40 (25.8%)
Plants' details	1 (1.5%)	13 (14.9%)	14 (9.0%)
Creatures	8 (11.8%)	12 (13.8%)	20 (12.9%)
Tablet	13 (19.1%)	22 (25.3%)	35 (22.6%)
Landscape	8 (11.8%)	6 (6.9%)	14 (9.0%)
Others	18 (26.5%)	14 (16.1%)	32 (20.6%)
Total	68	87	155

Note: $X^2_{(5)} = 12.04$, $^*p < 0.05$, $\varphi = 0.28$.

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FIGURE 8 A drawing example by a student in the experimental group (A20).

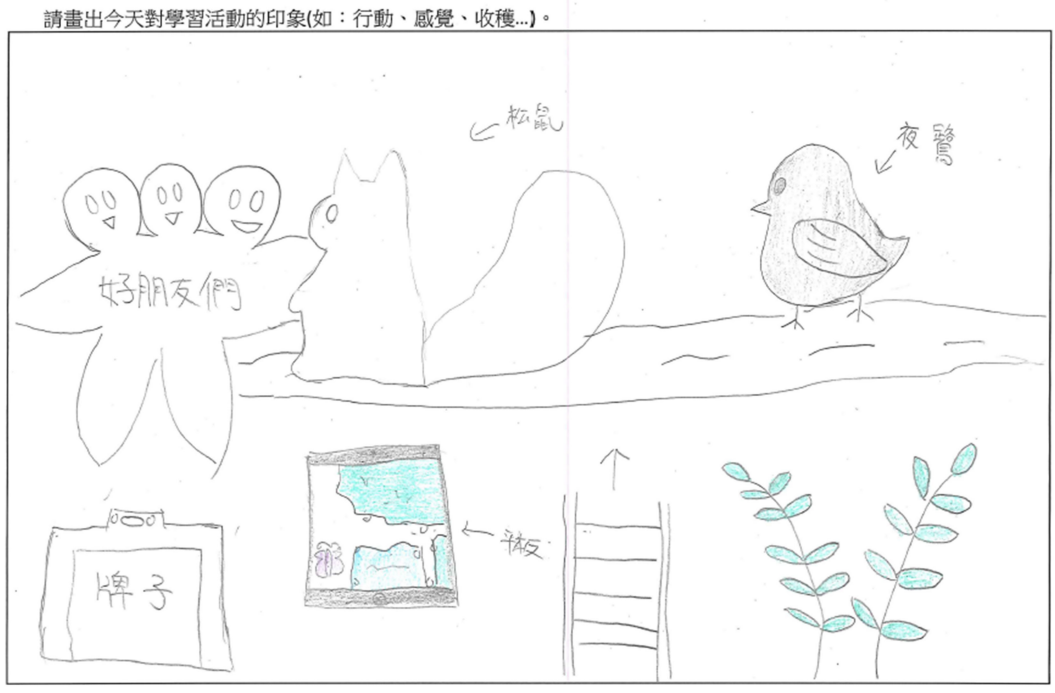


FIGURE 9 A drawing example by a student in the control group (C21).

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TABLE 6 The students' main emotions in the learning activities.

Emotion categories	Experimental group (intergroup ratio)	Control group (intergroup ratio)	Total (total ratio)
Joy	9 (45.0%)	21 (84.0%)	30 (66.7%)
Sadness	1 (5.0%)	1 (4.0%)	2 (4.4%)
Confusion	3 (15.0%)	0 (0.0%)	3 (6.7%)
Others	4 (20.0%)	0 (0.0%)	4 (8.9%)
No expression	3 (15.0%)	3 (12.0%)	6 (13.3%)
Total	20 (100%)	25 (100%)	45 (100%)

Note: X^2 (4) =11.39, *$p < 0.05$, $\varphi = 0.50$.

TABLE 7 The Mann–Whitney *U* test results of emotional engagement.

Group	*N*	Mean	SD	*U*	*z*	p^2	*r*
Experimental	17	4.68	0.51	95.00	−2.67**	0.01	0.43
Control	22	4.26	0.66				

**$p < 0.01$.

TABLE 8 The Mann–Whitney *U* test results of stay time (second) for each exhibit.

Group	*N*	Mean	*SD*	*U*	*z*	p^2	*r*
Experimental	15	674.23	330.13	56.00	−3.37***	0.001	0.55
Control	22	346.20	197.16				

***$p < 0.001$.

Behavioural engagement

Individual participation

The study collected all students' system logs for behaviour engagement analysis. Two students in the experimental group had no records; therefore, the behaviour engagement only included the data of 22 students in the control group and 15 students in the experimental group. A Mann–Whitney *U* test was conducted to evaluate the students' stay time for each target exhibit. As Table 8 shows, the students in the experimental group spent significantly more time on each exhibit than those in the control group ($U = 56.00$, $p < 0.001$).

Table 9 shows the equality of the participation results for the two groups. The Chi-square results showed that the control group included more free riders than the experimental group, especially for the second quest ($X^2_{(1)} = 13.75$, $p < 0.001$) and the sixth quest ($X^2_{(1)} = 10.42$, $p < 0.001$).

Problem-solving behaviours

To further identify the differences in the problem-solving behaviours of the two groups, 39 sets of code strings with 7519 behavioural codes were obtained for the follow-up sequential analysis. As shown in Table 10, the chi-square results showed that there was a significant difference in the behaviour frequency of the two groups ($X^2_{(11)} = 281.11$, $p < 0.001$). The

TABLE 9 The equality of participation results for each ARG quest.

Quest	Group	*N*	Free rider (intergroup ratio)	Engaged student (intergroup ratio)
Warm-up	Experimental	15	0 (0.0%)	15 (100.0%)
	Control	22	1 (4.5%)	21 (95.5%)
Quest 1	Experimental	15	0 (0.0%)	15 (100.0%)
	Control	22	3 (13.6%)	19 (86.4%)
Quest 2	Experimental	15	1 (6.7%)	14 (93.3%)
	Control	22	15 (68.2%)	7 (31.8%)
Quest 3	Experimental	15	3 (20.0%)	12 (80.0%)
	Control	22	4 (18.2%)	18 (81.8%)
Quest 4	Experimental	15	11 (73.3%)	4 (26.7%)
	Control	22	15 (68.2%)	7 (31.8%)
Quest 5	Experimental	15	4 (26.7%)	11 (73.3%)
	Control	22	12 (54.5%)	10 (45.5%)
Quest 6	Experimental	15	1 (6.7%)	14 (93.3%)
	Control	22	13 (59.1%)	9 (40.9%)

TABLE 10 The frequency and rate of behaviours of the two groups.

Behavioural code	Experimental group (intergroup ratio)	Control group (intergroup ratio)	Total (total ratio)
K (Skip the story)	142 (3.1%)	134 (4.5%)	276 (3.7%)
Q (Start a quest)	302 (6.7%)	399 (13.4%)	701 (9.3%)
F (Call for Bubi)	149 (3.3%)	35 (1.2%)	184 (2.4%)
H (Step-by-step guide)	79 (1.7%)	67 (2.2%)	146 (1.9%)
A (Chat with Bubi)	94 (2.1%)	0 (0%)	94 (1.3%)
P (Draw a note)	91 (2.0%)	54 (1.8%)	145 (1.9%)
S (Submit an answer)	502 (11.1%)	467 (15.6%)	969 (12.9%)
W (Wrong answer)	1572 (34.7%)	849 (28.4%)	2421 (32.2%)
C (Correct answer)	205 (4.5%)	174 (5.8%)	379 (5.0%)
L (Read quest clues)	132 (2.9%)	94 (3.1%)	226 (3.0%)
N (Read learning summaries)	204 (4.5%)	174 (5.8%)	378 (5.0%)
M (Explore map)	1061 (23.4%)	539 (18.1%)	1600 (21.3%)
Total	4533 (100%)	2986 (100%)	7519 (100%)

Note: $X^2_{(11)} = 2\ 81.11$, *** $p < 0.001$, $\varphi = 0.19$.

students who learned with the AIC-ARG had 4533 behavioural codes, which is significantly more than the 2986 behavioural codes of those who learned with the C-ARG.

Tables 11 and 12 show the significant sequences of sequential analysis results, which indicated that there were 21 and 26 sets of significant behavioural sequences in the control and experimental groups respectively. Figures 10 and 11 display the comparison of the significant behavioural sequences of the two groups, which indicates that there were 16 similar behavioural sequences. The results indicate that the students in both groups repeated resizing the exploring map (M → M) to locate the target exhibits. Besides, most students tended to skip the stories (K → Q) and then started to answer the quest without reading the

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TABLE 11 Adjusted residuals table of the control group.

Code	C	F	S	K	L	M	N	P	Q	W	H
C	−3.31	−1.45	−5.80	−2.79	−2.46	−6.07	53.83*	−1.80	−5.06	−8.65	−2.06
F	0.77	−0.64	−2.54	−0.36	8.62*	−1.40	−1.45	−0.79	−0.65	−2.65	11.71*
S	6.61*	5.04*	−10.15	22.80*	4.53*	−6.76	−5.79	8.38*	1.26	−3.68	2.46*
K	9.11*	1.27	−5.03	−1.07	1.41	−1.47	−2.87	0.49	7.31*	−3.16	−1.79
L	−2.46	−1.08	20.14*	−1.56	−1.25	−2.51	−2.46	−1.34	−3.77	−6.44	−1.53
M	−4.45	−1.86	−10.00	−4.11	−3.07	41.52*	−6.10	−1.97	−3.70	−13.95	−2.97
N	−1.82	−1.41	−5.64	−2.30	−1.94	−3.56	−3.22	−1.13	29.00*	−8.05	−1.46
P	−1.80	−0.79	16.79*	−1.52	−1.34	−3.41	−1.80	−0.98	−2.75	−4.70	−1.12
Q	−5.07	−2.22	39.69*	−3.99	−3.77	−8.90	−5.07	−2.76	−0.34	−13.25	−3.16
W	2.30*	1.13	−15.17	−7.09	2.67*	−14.20	−8.65	−0.40	−9.56	34.88*	4.03*
H	−1.01	−0.90	−3.62	−1.74	−1.53	−3.59	−2.06	−1.12	−2.41	11.57*	−1.29

*z scores > 1.96 ($p < 0.05$).

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TABLE 12 Adjusted residuals table of the experimental group.

	A	C	F	S	K	L	M	N	P	Q	W	H
A	−1.43	4.86*	2.27*	−3.14	−1.04	4.48*	−3.44	−2.13	3.04*	−0.92	−2.36	15.37*
C	−2.14	−3.20	−2.71	−5.19	−2.50	−2.55	−7.93	66.60*	−2.10	−3.61	−10.73	−1.96
F	53.16*	−1.10	−1.35	−4.12	−2.11	0.82	−5.84	−2.69	−1.18	−1.94	−8.00	14.27*
S	−3.45	5.35*	1.49	−8.36	28.76*	6.34*	−7.64	−5.14	13.19*	0.25	−4.20	2.29*
K	−1.76	4.35*	1.60	−4.27	−1.02	1.46	1.22	−2.63	1.31	9.55*	−4.17	−1.61
L	−1.70	−2.55	−1.66	29.56*	−1.99	−0.98	−4.98	−2.54	−1.67	−2.39	−8.54	−1.56
M	−5.41	−3.36	−4.69	−12.34	−5.68	−3.94	52.31*	−7.90	−3.07	0.51	−24.92	−3.88
N	−2.08	3.58*	3.11*	−5.04	−0.21	−0.73	−1.62	−3.10	−0.48	28.16*	−9.65	−1.90
P	−1.41	−1.60	−1.78	26.54*	−1.64	−1.67	−5.32	−2.10	−1.38	−2.13	−7.06	−1.29
Q	−2.62	−3.92	−2.32	47.49*	−3.06	−3.12	−9.36	−3.91	−2.58	−2.37	−13.15	−2.40
W	−7.17	−0.97	4.90*	−17.39	−7.99	1.48	−24.98	−10.70	−3.05	−10.28	48.46*	−3.23
H	−1.31	4.03*	−1.02	−3.18	−1.53	−1.56	−4.68	−1.95	−1.29	−1.92	8.91*	−1.20

*z scores > 1.96 ($p < 0.05$).

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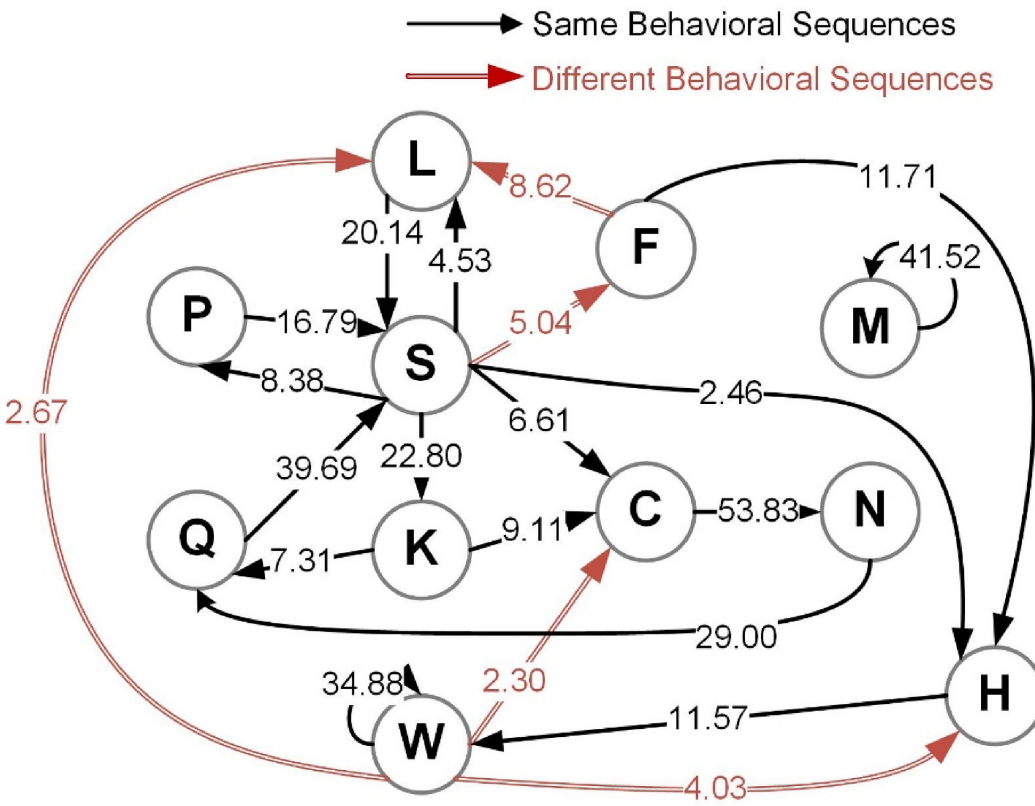


FIGURE 10 The sequential behavioural patterns of the control group.

supplementary clues (Q → S). The two-way sequence between S and L indicates that many students read the clues (L) after they started to answer (S), and they reviewed the clues to help them identify the answer choices. The two-way sequence between P and S indicates that the students usually took notes (P) to confirm their answers (S). Besides, they may read the stories again to find related clues after they start to answer (S → K).

Figure 12 indicates the differences in the problem-solving behaviours of the two groups. The results show that the students in the two groups usually used trial and error to solve problems after they submitted wrong answers (W → W). Regarding the students who learned with the C-ARG, they read the clues (W → L) or revised the hints (W → H, H → W) to solve the problems. To investigate how the students in the control group solved the problems successfully, there are three significant behavioural sequences: W → C, S → C and K → C. The sequence W → C reveals that most students in the control group solved problems with trial and error. The sequence S → C indicates that the students solved the problem successfully. However, the sequence K → C implied free-riding behaviours so that the students solved the problem at once, and skipped the stories.

Regarding the students who learned with the AIC-ARG, the sequence W → F indicates that the students in the experimental group sought Bubi's help while encountering problems. The bidirectional sequence (F → A, A → F) indicates that the students in the experimental group usually worked with the AI-based chatbot to solve the problems. The sequence A → C presents that the AI-based chatbot effectively supported the students in solving problems. The students in the experimental group also read the hints (H → C) to solve the problems.

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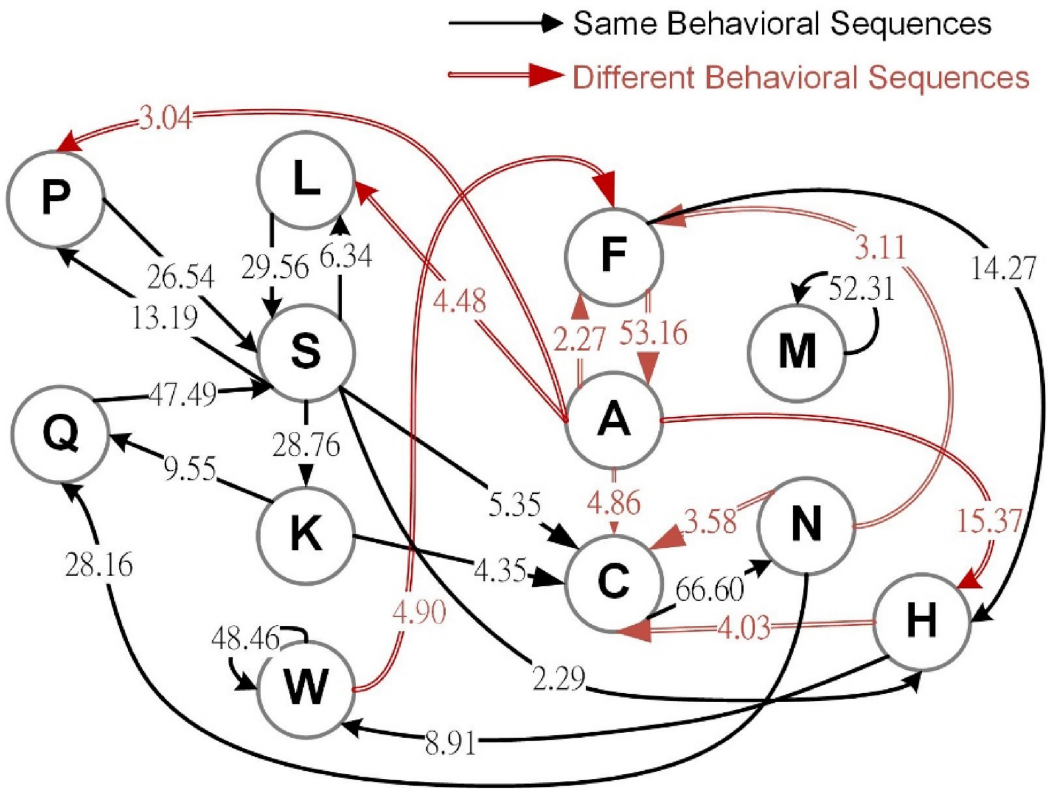


FIGURE 11 The sequential behavioural patterns of the experimental group.

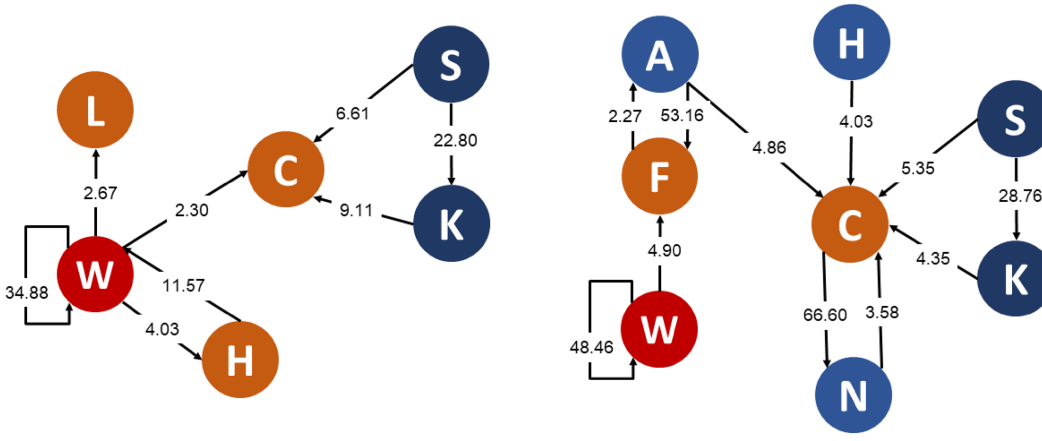


FIGURE 12 The sequential pattern of problem-solving behaviours in the control group (left) and experimental group (right).

Besides, the bidirectional sequence (C→N, N→C) demonstrates that the students in the experimental group learned the related knowledge and thus built the required knowledge to solve the problems. What is worth mentioning is that W→C is not seen in the experimental group. The results of significant sequences revealed that students in the experimental group

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applied more problem-solving strategies rather than trial and error, which they may have learned from their interactions with the AI-based chatbot and from the collected information, such as hints and learning summaries.

DISCUSSION AND CONCLUSIONS

This study explored the effects of AI-based chatbots on students' learning outcomes in a museum. The results showed that there was a medium effect of AI-based chatbots on the students' metacognitive awareness and engagement. However, the students who learned with the AIC-ARG and C-ARG performed similarly in terms of their learning achievements.

Regarding the effect of the AIC-ARG approach on engagement, the results found that the students in both groups presented different cognitive foci in the learning activities. The students who learned with the AIC-ARG were seen to engage more in problem-solving activities, while those who learned with the C-ARG focused more on learning the characteristics of plants. The pull-and-push motivation theory explained that task difficulty is the key to whether students continue engaging in learning activities (Dean & Suhartanto, 2019). The perceived challenges of learning goals could affect students' willingness to continuously make efforts to achieve the goals (Tornqvist & Tichon, 2021). Considering that the C-ARG's feedback may fail to support students in solving problems effectively, the frustration they experience in problem solving can influence their motivation to continue engaging in the learning activities (Whitton et al., 2014). Besides, the pretest may give the students the wrong idea that the learning goal is to gain better learning achievement in the posttest. Hence, most students who learned with the C-ARG paid more attention to the plant details instead of the problem-solving activities.

This study also found that the students who learned with the AIC-ARG had better emotional engagement than those who learned with the C-ARG. Previous research explained that AI-based chatbots encouraged students' social interactions, and facilitated their emotional engagement (Gaia et al., 2019). The drawing analysis results described that the students who learned with the AIC-ARG had richer emotional experiences, such as confusion or other emotions. Flow theory describes that students who put more effort into engagement express highly immersed feelings, such as being excited or nervous (Ninaus et al., 2019). Their higher flow experiences may explain why the students who learned with the AIC-ARG had better persistence in problem-solving tasks (Kiili et al., 2014). The behavioural engagement results showed that the students who learned with the AIC-ARG had significantly more staying time for each exhibit, and better behavioural engagement than those who learned with the C-ARG. Besides, the AI-based chatbot encouraged students' individual participation, which improved the equality of participation in collaborative learning. This may be because the AI-based chatbot provided each student with adaptive feedback to facilitate their engagement in the learning activities (Winkler et al., 2021). The interactions between chatbots and students improved students' responsibility for promoting individual contributions and interactions among team members; therefore, it benefited collaborative learning to avoid unequal participation (Yilmaz & Yilmaz, 2019).

The behavioural sequential analysis results also indicated that the students who learned with the C-ARG presented single-loop learning behaviours, and they usually used the simple problem-solving strategy, trial and error, to solve encountered problems. Conversely, the AI-based chatbot facilitated students' collection of information and application of advanced problem-solving strategies to achieve double-loop learning (ie, acquiring new knowledge or information before making any attempt to solve problems). Previous research explained that only appropriate learning supports benefit students' self-regulated learning (Kiili, 2007). Researchers have indicated that corrective feedback may fail to support students' reflections

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to achieve double-loop learning, unless the students can interpret the meanings of the given feedback (Carless, 2019). This may explain why the C-ARG hardly facilitated double-loop learning because the given feedback was not helpful for the students. However, the AI-based chatbot provided more details when the students were unable to interpret the given feedback. Thus, the AI-based chatbot effectively facilitated double-loop learning to achieve successful problem solving.

To sum up, the AIC-ARG approach has the potential to promote students' metacognitive awareness and engagement with adaptive feedback. In particular, the recent development of generative AI-based chatbots (eg, ChatGPT) has further provided opportunities for using AI-based chatbots as a smart learning partner in diverse learning activities to support students in achieving double-loop learning with immediate and adaptive feedback, as suggested by several scholars (eg, Tlili et al., 2023).

However, there are some limitations in this study that should be noted. First, due to COVID-19 affecting schools' willingness to participate in field trip learning, the small sample size may have affected the validity of the inference of the study results. Considering that small sample sizes influence the reliability of studies, the results may differ in similar conditions (Button et al., 2013). Thus, there is a need to examine the effects of the AIC-ARG in future studies. A pre-registration of studies and calculation of study size can be used to improve the reliability of study results. Second, this study only examined the learning effectiveness of school groups on a museum field trip. Considering that different visitor groups may have different preferences and visiting behaviours, other visitor groups, such as families or adult visitors, may perform differently. Third, this study was a short-term intervention experiment of a half-day museum learning activity. Considering the novelty effect of technology and the influences of familiarity with AI-based chatbots, future research can conduct a long-term study to examine the learning effectiveness. Besides, delayed posttests or a long-term longitudinal tracking study can be conducted to explore the long-term effects of AI-based chatbots on learning.

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CONFLICT OF INTEREST STATEMENT

There is no potential conflict of interest in this study.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The participants were protected by hiding their personal information during the research process. They knew that the participation was voluntary and they could withdraw from the study at any time.

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